4 Multiple Integrals

4.2 Double Integrals on General Regions

Definition 4.2.1 Let $f(x,y)$ be a function defined on D, which is bounded. Let R be a rectangle which contains D. Define the function

$$F(x,y) = \begin{cases} f(x,y) & (x,y) \in D \\ 0 & (x,y) \notin D \end{cases}$$

We then define the integral of f over D as

$$\iint_D f(x,y)\,dA = \iint_R F(x,y)\,dA.$$

Example 4.2.2 Find the volume of the solid that lies under the paraboloid $z = x^2 + y^2$ and above the region D which is given by

$$D = \{(x,y) : 0 \le x \le 2, 0 \le y \le x\}.$$

Example 4.2.3 Find the volume of the solid which lies between the xy-plane and the surface

$$z = 4 - x - y,$$

and lies above the region

$$D = \{(x, y) : y^2 - 1 \leq x \leq 1 - y^2\}$$

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Example 4.2.4 Evaluate

$$\iint_D xy \, dA,$$

where D is the region bounded by the line $y = x - 1$ and the parabola $y^2 = 2x + 6$.

4.2.1 Changing the Order of Integration

Example 4.2.5 Evaluate the integral

$$\int_0^1 \int_x^1 \sin(y^2) dy dx.$$

Example 4.2.6 Evaluate the integral

$$\int_0^1 \int_{3y}^3 e^{x^2}\, dxdy.$$

Theorem 4.2.7 *(Properties of Double Integrals)*

1. $\iint_D (f_1(x,y) + f_2(x,y))\, dA = \iint_D f_1(x,y)\, dA + \iint_D f_2(x,y)\, DA.$
2. $\iint_D cf(x,y)\, dA = c \iint_D f(x,y)\, dA.$
3. $\iint_D f(x,y)\, dA = \iint_{D_1} f(x,y)\, dA + \iint_{D_2} f(x,y)\, dA$, *where* $D = D_1 \cup D_2$ *and* D_1 *and* D_2 *do not overlap, except possibly on their boundaries.*
4. $\iint_D 1\, dA = A(D).$
5. *If* $m \leq f(x,y) \leq M$ *on* D, *then* $mA(D) \leq \iint_D f(x,y)\, dA \leq MA(D).$